

FINAL EXPLANATION: BIOLOGY GRADE 10

Student Assessment Document

FINAL EXPLANATION: BIOLOGY GRADE 10 – SUB-STRAND 1.1 CELL STRUCTURE: FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Driving question: Why do cells in our body look so different from each other, and how does the shape of a cell determine the job it can do? Use everything you learned in Lessons 1 to 12 to write a complete, evidence-based explanation of The Healing Cut: the Form 1 student in Nairobi who cut a finger while slicing onions for supper. Answer each section in the space under its prompt. Write in full sentences, use correct scientific words, and refer to evidence from our lessons (diagrams, the onion and cheek cell practicals, the specialised cell studies, the sickle-cell micrographs and the cell organisation hierarchy). Where a question asks you to make a claim, give your Claim, your Evidence and your Reasoning. Sketches and labelled diagrams are welcome.

Section 1: Cells as the Starting Point

Think about the cut finger. (a) State the three tenets of cell theory and name the scientist linked to the third tenet. (b) Use cell theory to explain where the new skin cells came from as the finger healed. (c) The bacteria that might infect the cut (for example *E. coli* from badly prepared nyama choma) are prokaryotes, while the body cells fighting them are eukaryotes. Describe two structural differences between these two cell types.

Write your answer here:

Section 2: Plant Cells and Animal Cells

In Lessons 4 to 7 you studied plant cells and animal cells, including onion epidermal cells and your own cheek cells under the microscope. (a) Name the structures that plant cells have but animal cells lack, and the structures animal cells have that plant cells lack. (b) What did you observe in the onion cells that was NOT there in your cheek cells, and what did you see in both? (c) You did not see chloroplasts in the onion epidermal cells. Explain why this is not a mistake. (d) Explain why the lack of a cell wall in animal cells is useful for the cells that close the cut finger.

Write your answer here:

Section 3: Specialised Cells and the Healing Cut

Healing the cut involves platelets (cell fragments), neutrophils, macrophages, fibroblasts and epidermal cells. (a) Choose THREE of these. For each, describe one structural feature and explain how it suits the job it does in the healing sequence. (b) Use the sickle-cell anaemia micrographs from Lesson 8 as evidence to write a Claim, Evidence and Reasoning paragraph answering: Does the shape of a cell affect the job it can do? (c) Choose one specialised plant cell (such as a root hair cell, xylem vessel or palisade cell) and explain how its structure suits its function.

Write your answer here:

Section 4: Same DNA, Different Cells

The puzzle that started our unit was: if all cells in one organism carry the same DNA, why do they look so different? (a) Use what you learned to answer this puzzle. (b) Use your evidence from the onion and cheek cell practicals, or the specialised cells you studied, to support your answer to (a). (c) Using the cell organisation hierarchy, explain why no single cell, such as a platelet or a fibroblast, can heal the cut alone.

Write your answer here:

Section 5: Final Explanation of The Healing Cut

Write a complete explanation answering the driving question: Why do cells in our body look so different from each other, and how does the shape of a cell determine the job it can do? Tell the story of the cut finger in order, from the moment of the cut until the skin is closed. Name at least four cell types or cell fragments involved, describe the shape or organelle that suits each to its job, and show how they work together. Finish by saying how your thinking has changed since Lesson 1.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Use of cell structure knowledge	Accurately names and describes organelles and structures (cell theory, prokaryote versus eukaryote, plant versus animal cells) and links each to its function with correct scientific vocabulary.	Names most structures and functions correctly, with minor errors or gaps in the comparisons.	Names few structures, or confuses plant and animal features or prokaryotes and eukaryotes.
Linking structure to function	For several specialised cells, clearly explains how a specific structural feature enables the job the cell does, including the sickle-cell evidence.	Explains structure-function links for some cells but the reasoning is sometimes incomplete.	Lists features or functions without connecting them to each other.
Use of evidence and reasoning (CER)	Makes clear claims supported by specific evidence from lessons (onion and cheek practicals, micrographs) and reasons logically from evidence to claim.	Gives claims with some evidence, but the reasoning is thin or one part of CER is missing.	Makes statements with little or no evidence from the lessons.

Explaining the healing cut and the driving question	Gives a complete, correctly sequenced story of healing, naming at least four cell types or fragments, explaining same DNA with different gene expression, and the tissue-to-organism hierarchy.	Gives a mostly complete explanation, with some cell types, the sequence or the DNA and hierarchy ideas weak.	Gives a partial or unordered explanation that does not answer the driving question.
Reflection and communication	Clearly shows how thinking has changed since Lesson 1, written in organised, clear sentences with correct terms.	Shows some change in thinking, with generally clear writing.	Reflection is missing or very general, and the writing is hard to follow.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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